

# Business Entity Resolution

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## Problem and design goal

For each deduplicated Source 1 business, predict all matching Source 2 and Source 3 IDs, including an empty list for singletons. Training has 2.21 million Source 1 records, 10.32 million targets, and 7.64 million labeled links. Test adds France with no French training labels. The metric is macro F0.5 over Source 1 groups, so false merges and missed singletons matter alongside retrieval recall.

### 1. Retrieve candidates with complementary routes

A local DuckDB index stores Unicode name and address tokens, target frequencies, and compact legal-suffix name keys. Each query uses rare name and address token postings within the same country. Separate name and address Jaro-Winkler top-100 quotas protect matches when either field is corrupted or absent. Compact and accent-folded name equality rescue punctuation and French accents. For India, an additional top-50 address-token-overlap channel helps cross-script names. Country is an open-set string, so France is processed without hard-coded exclusion.

### 2. Learned last-stage block, then final matching

A frozen 23-feature LightGBM ranks the broad union and passes at most 40 candidates per Source 1. A local word TF-IDF search adds up to 20 IDs per India query. For US queries with at most three initial matches, a second exact sparse TF-IDF address search adds up to five IDs. A character 3-4-gram name TF-IDF search adds up to ten IDs for US queries with at most two initial matches. A US character 3-4-gram address search shortlists on 16 strong terms, reranks 300 address keys, and adds up to ten IDs per US query. The recorded final candidate TSV is the union actually scored. Top 40 retained 70,735/70,741 broad-reachable true links in development; the India quota added 1,346 reachable links. No pretrained weights or external data are used for these searches.

A separate 35-feature Unicode/transliteration LightGBM was trained on 5.25 million pairs mined from 190,086 training-owned queries. A 36-feature specialist replaces its score for targets with empty addresses, using target core-name frequency. India/other thresholds are 0.65/0.75; blank-address threshold is 0.80.

### 3. Resolve group and ownership conflicts

Rank selected links by model probability and retain at most 11 per Source 1, the largest true group in the full training labels. Assign a target predicted by multiple Source 1 records to its highest-scoring owner. Both passes preserve the exact candidate subset contract. A fixed address-number veto was rejected for this model because it reduced labeled F0.5.

## Retrieval checks that changed the design

- The broad lexical union had 4.74 million development pairs and an oracle macro F0.5 ceiling of 0.96793. The learned top-40 block kept the ceiling at 0.96788.
- A full-index India address TF-IDF search found 1,870/3,070 missed true links among its top 100 address keys. Its bounded quota improved measured end-to-end F0.5 on development and separate frozen validation.
- The gated US top-five address route added 28 true links and zero false links on frozen validation; the same final matcher scored every added pair.
- The gated US top-ten character-name route added another seven true links and zero false links after the US address route on frozen validation.
- The US top-ten character-address route added 36 true and three false links after the name route on frozen validation.

# Measured evidence and submission

All scores below are local macro F0.5; public and private leaderboard scores must be measured separately.

## Held-out evaluation

| Check                        | Scope                             | Result              |
|------------------------------|-----------------------------------|---------------------|
| Combined candidate oracle    | 22,133 development queries        | 0.96793 upper bound |
| Learned top-40 oracle        | 22,133 development queries        | 0.96788 upper bound |
| Original 23-feature baseline | Full development, 3 output passes | 0.88331             |
| Hard + blank specialist      | Development, top 40, cap + owner  | 0.92322             |
| Plus India address top 20    | Development, cap + owner          | 0.93183             |
| Hard + blank specialist      | Frozen 1,994-query validation     | 0.91845             |
| Plus India address top 20    | Frozen 1,994-query validation     | 0.92752             |
| Plus gated US address top 5  | Frozen 1,994-query validation     | 0.92989             |
| Plus gated US name top 10    | Frozen 1,994-query validation     | 0.93118             |
| Plus US char address top 10  | Frozen 1,994-query validation     | 0.93263             |

## What the measurements mean

- The candidate oracle assumes perfect scoring on retrieved pairs. It measures the retrieval ceiling, not an achieved model score.
- The learned top-40 block controls most candidate volume. India address top 20 raised frozen macro F0.5 by 0.00907; gated US address top 5 added 0.00237. Gated US character-name top 10 added 0.00129. US character-address top 10 then added 0.00145. Every quota was measured before full test inference.
- The general matcher compares original Unicode and local ASCII-transliterated name/address views, plus soft number agreement. The blank-address specialist uses locally computed core-name rarity. No French labels guided either model.
- France has no truth labels. In a 20,000-query test slice, generic-name collisions remained before capping. The top-11 cap was retained; no French F0.5 is claimed.

## Reproducibility and fair play

The final ZIP contains the exact last-stage candidate TSV, matching TSV, runnable Python source, four locally trained model weights, pinned dependencies, and the full methodology. Both broad lexical and TF-IDF indexes are reproducible intermediates from supplied records. Every predicted ID belongs to the submitted candidate set. Streaming and official validators are run before submission. No supplied record text is sent to a remote service.

## Team

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